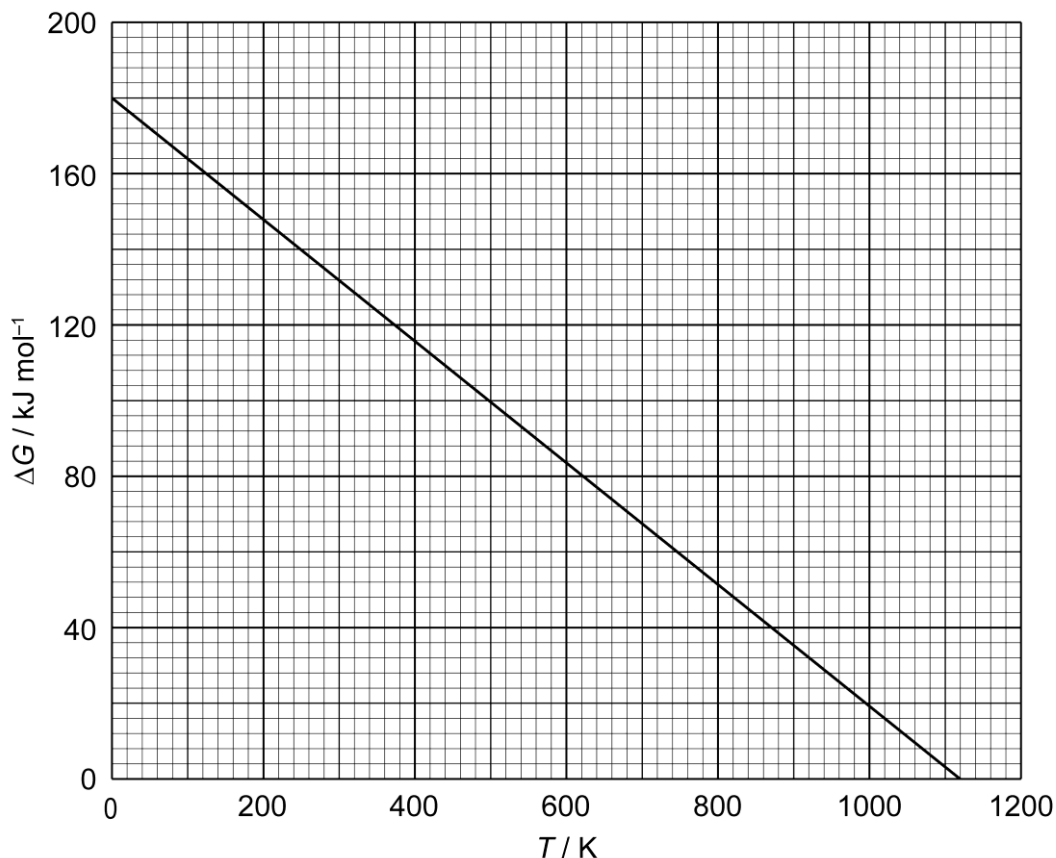


- 12 When heated, magnesium carbonate decomposes to form carbon dioxide and magnesium oxide.

A graphical plot of ΔG versus T , describing the change of the Gibbs free energy of the decomposition of magnesium carbonate with respect to temperature, is shown below.



Using the information from the graph, what is the value of $\Delta S^\ominus$ for the decomposition reaction?

- A** $+6.04 \times 10^2 \text{ J mol}^{-1} \text{ K}^{-1}$ **B** $-6.04 \times 10^2 \text{ J mol}^{-1} \text{ K}^{-1}$
C $+1.61 \times 10^2 \text{ J mol}^{-1} \text{ K}^{-1}$ **D** $-1.61 \times 10^2 \text{ J mol}^{-1} \text{ K}^{-1}$